

Reconcile Once, Write Anytime: A Trust-Tiered Librarian and a Multi-Agent Writer for Drift-Free, Point-in-Time Research

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Abstract

Long-form research reports generated by large language models (LLMs) drift, contradict themselves, and lose provenance: the same metric appears with different values in different sections, numbers arrive without sources, and rumor is quoted as confidently as an audited filing. We present a deployment-oriented two-tier agentic system that separates a *maintained, point-in-time knowledge library* from *report writing*. A deterministic “librarian” continuously ingests public, timestamped sources into a trust-tiered *ontology*, layering evidence cards, an authoritative metric ledger, and a claim graph into an always-current source of truth, not per-query RAG over raw chunks. A portable multi-agent “writer” runtime then composes a long, contradiction-free, evidence-grounded report at any knowledge cutoff T , reading only evidence with $as_of \leq T$ (no look-ahead); red-team verdicts flow back into the librarian, closing the loop. We evaluate on a self-collected, production-scale public corpus of **6,130 sources** yielding **555,926 evidence cards** (SEC EDGAR filings across 295 issuers and 11 sectors, U.S. Bureau of Labor Statistics macro releases, and Wikipedia). From the *one* library we compose four flagship point-in-time reports on distinct theses (AI-compute, energy, healthcare/pharma, banks) and run eight mechanical, reproducible experiments, whose headline metrics come from a deterministic quality-control (QC) gate, itself validated by defect-injection meta-evaluation at recall 1.0 and precision 1.0 against negative controls. A shared metric ledger removes 6,845 cross-section figure contradictions to zero. Tier-first selection is correct on 22/22 gold cases where a popularity-first baseline scores only 9/22; trust tiering leaks zero media-sourced numbers into hard evidence, and no government statistic displaces a company’s own filing. A red-team refutation propagates back through a source override and self-corrects a later run with zero manual edits. Point-in-time replay exhibits zero look-ahead violations across seven cutoffs while the library grows from 235,373 to 555,312 cards. Finally, difficulty-tiered model routing with bounded parallelism *exceeds* the all-Opus quality ceiling on a graded score while running 3.7× faster than serial, with cost and latency recorded on every run.

Keywords

distributed multi-agent systems, agentic AI, parallel orchestration, model routing, retrieval, report generation, trust tiering, point-in-time evaluation, deployed systems

1 Introduction

Automated long-form research (equity notes, market landscapes, due-diligence memos) is a natural target for LLM agents, yet naive generation fails in ways that matter precisely where the stakes are highest. Three failure modes recur. (i) *Numeric drift*: a report cites a

company’s capital expenditure as one figure in the revenue section and a different figure in the capital-intensity section, because each paragraph was grounded independently. (ii) *Provenance loss*: numbers appear with no traceable source, and the reader cannot tell an audited disclosure from a fluent hallucination [13]. (iii) *Trust flattening*: a pre-launch rumor and a filed 10-Q are quoted with equal confidence, because the system has no notion of source authority.

Retrieval-augmented generation (RAG) [10, 16] re-retrieves raw chunks per query and grounds each answer in isolation, maintaining no *consistent, evolving* body of judgements, so contradictions and stale figures recur at every generation; agentic writers [11, 24] coordinate sections but inherit the same gaps. We argue the fix is architectural: reconcile evidence *once* into a maintained ontology (trust-ranked, timestamped, incrementally updated), separated from the *writing* of any report, so consistency and provenance are properties of the store, governed by source authority rather than popularity.

We built such a system, validated it internally on a private corpus, and evaluate it here on a *production-scale* public corpus (6,130 sources and 555,926 evidence cards across 295 issuers and 11 sectors, from which we compose four flagship reports on distinct theses), as an **industry case study with end-to-end reproducible results**, not a state-of-the-art (SOTA) claim. Our contributions:

- (1) **A coupled two-phase system** (§2): a deterministic librarian maintains a timestamped knowledge ontology, so new filings incrementally refresh the source of truth without re-indexing (Phase A); and a portable, model-routed multi-agent runtime generates a point-in-time report at cutoff T (Phase B).
- (2) **A distributed multi-agent writer with a shared-store coordination substrate** (§2.1): heterogeneous agents (per-section composers and an adversarial red-team prosecutor) run under bounded-concurrency parallelism and difficulty-tiered model routing, coordinated *indirectly* through the trust-tiered store rather than by direct message passing, so concurrency never costs cross-section consistency.
- (3) **Trust-tiered consistency mechanisms as deployed governance** (§3): an official-first metric ledger that ranks source *authority* above popularity (tier $\rightarrow$ corroboration $\rightarrow$ recency), a typed claim graph, and a six-check deterministic QC gate that *blocks* delivery and is itself meta-validated by defect injection (E5), not a post-hoc score.
- (4) **A self-correcting write-back loop** (§2): a report-side red-team refutation flows back as a source override that self-corrects the ledger at the next cutoff with no manual edits, and idempotent regeneration flags any promoted claim whose approved evidence silently vanishes (an “anchor swap”) for re-validation, so the library only ever improves.
- (5) **A mechanical, meta-eval-validated evaluation** (§5) others can reuse, including the distributed-execution cost/latency of

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routing and bounded parallelism (recorded on every run) and a *temporal point-in-time* result the living library uniquely enables.

2 System

The system runs as two decoupled phases coupled by a timestamped store and closed by a write-back loop (Fig. 1).

Phase A: the librarian (deterministic). A pipeline of deterministic Python stages ingests public sources, recording each one’s true publication date and trust tier. The librarian maintains knowledge at three progressively-distilled layers: quote-grounded *evidence cards* (raw facts, each pinned to a source quote), an *authoritative metric ledger* (one reconciled value per company-metric pair), and a *claim graph* of contradicts, supersedes, and qualifies edges over those values. Each layer is derived deterministically from the one below, so a number traces down to its source quote and up to its conflicts. The core costs essentially nothing to re-run; an LLM is used only at one clearly-bounded seam: a cheap refinement pass (Claude Haiku 4.5) that corrects a numeric card’s value/unit against its own quote or demotes it to qualitative. The headline metrics, computed deterministically, are thus reproducible given a fixed store. Appendix A traces one real metric through all three layers and into a delivered report; Appendix C gives their required fields.

The bridge: point-in-time projection. Given a cutoff T , the bridge projects the store into the writer’s four artifacts (outline, evidence cards, metric ledger, claim graph), filtering to *as_of* $\leq T$. This is the no-look-ahead seam: a report “as of” a past date sees exactly the evidence that existed then. Object mapping is near 1:1 (both tiers speak the same three layers), so the bridge is a data adapter, not a rewrite.

Phase B: the portable writer runtime. The writer is a self-contained, headless runtime over a tiered LLM provider and a bounded-concurrency pool, with no dependency on an external agent framework, so it embeds equally in a service or a batch job. Its orchestration wraps a set of deterministic scripts (slice, tag-normalize, QC, render) around the LLM calls. The workflow is a fixed directed acyclic graph (DAG): slice each section $\rightarrow$ compose (one LLM call per section, tier-routed) $\rightarrow$ normalize $\rightarrow$ red-team (a Claude Opus 4.8 “prosecutor” per section that returns holds/weak/refuted verdicts) $\rightarrow$ apply verdicts $\rightarrow$ a bounded rewrite of affected sections $\rightarrow$ deterministic convergence backstop $\rightarrow$ QC gate $\rightarrow$ render. We use standard multi-agent building blocks (file-per-agent artifacts, contract-first outlines, tool-mediated deterministic checks) but implement them in a runtime of our own, without binding to any external framework [2].

Write-back loop. When the red-team refutes a card, the verdict maps to a librarian *source_override/claim* refuted, so the next run at the next cutoff inherits the correction: the store is a living library, not a static dump. The write-back is guarded: an override never invents a value, it only demotes the refuted source so the ledger falls back to a pre-existing same-kind alternative, and every change appends a row to an append-only audit log, so promotion is human-gated and reversible. Regeneration is idempotent: it carries a human promotion forward rather than resetting it, but snapshots the exact evidence approved, so if that evidence later vanishes,

even when back-filled cards keep the count unchanged (an “anchor swap”), the claim is flagged for re-validation, not silently kept. Algorithm 1 states the two phases end to end.

2.1 The distributed multi-agent design

Phase B is where the system is genuinely *distributed*, by deliberate design. **(1) Parallel per-section agents.** The report outline is a partition: each section is an independent compose task, and the sections fan out across a bounded-concurrency pool (§3) rather than being written serially. Sections are the natural unit of parallelism because the outline contract makes them near-independent: the only shared state is the metric ledger, read-only at compose time. **(2) Heterogeneous agents.** A difficulty router sends conflict-touching sections to a stronger, costlier model (Opus) and routine sections to a cheaper one (Sonnet), so compute is spent where the reasoning is hard: classic heterogeneous scheduling, applied to LLM agents. **(3) Separation of powers.** Composition and criticism are *different* agents with opposing objectives: a composer writes to satisfy the section contract, an independent Opus “prosecutor” red-teams the draft to break it. Neither grades its own work, and the arbiter that decides delivery is the deterministic QC gate, not an LLM. **(4) Coordination through a shared store, not messages.** The agents never talk to each other directly; they coordinate *stigmatically* through the trust-tiered store: composers read the same authoritative ledger, and the red-team writes verdicts back to it. This adds parallelism without the usual multi-agent failure mode of concurrent writers diverging: because the single authoritative value lives in the shared ledger rather than in each agent’s context, two sections physically cannot cite different numbers for the same metric. The distributed design thus buys throughput (§5, E6) *without* paying in consistency. Figure 3 (App. D) draws the four mechanisms as one picture.

Why the fan-out is safe. The consistency guarantee is structural, not a matter of scheduling. The bridge (§2) emits an *immutable, point-in-time snapshot* at cutoff T , and every composer reads from that single snapshot, so within one run there are no concurrent writers: no lock, barrier, or two-phase commit is needed, and the classic shared-memory races (write-write, torn reads, deadlock) cannot arise. The only writer is the red-team’s write-back, deferred to the *next* cutoff, never mid-run. Read-only fan-out over an immutable snapshot makes each compose step idempotent and order-independent, so the worker count K trades latency against cost (E6) but cannot change the delivered numbers.

3 Trust & Consistency Mechanisms

Source tiers and permitted use. Every source is typed from text/path cues and assigned a trust tier together with a *permitted use* that governs whether its numbers may be cited: U.S. Securities and Exchange Commission (SEC) filings become *official* (usable as hard evidence), U.S. Bureau of Labor Statistics (BLS) macro-statistics releases become *gov_stat* (supporting evidence: authoritative, but for macro context, not a company’s own figures), and Wikipedia becomes *media* (routing only). The tiers are strictly ordered (official $>$ gov_stat $>$ sell_side $>$ media; *sell_side* is analyst research, used in company deployments but omitted here as it is not redistributable, so it appears only in E4’s gold set): routing-only sources

A two-tier agentic system: maintain one library, write many point-in-time reports

A maintained trust-tiered store (Phase A), projected at a knowledge cutoff T , is composed into a contradiction-free, fully-cited report (Phase B); verdicts write back.

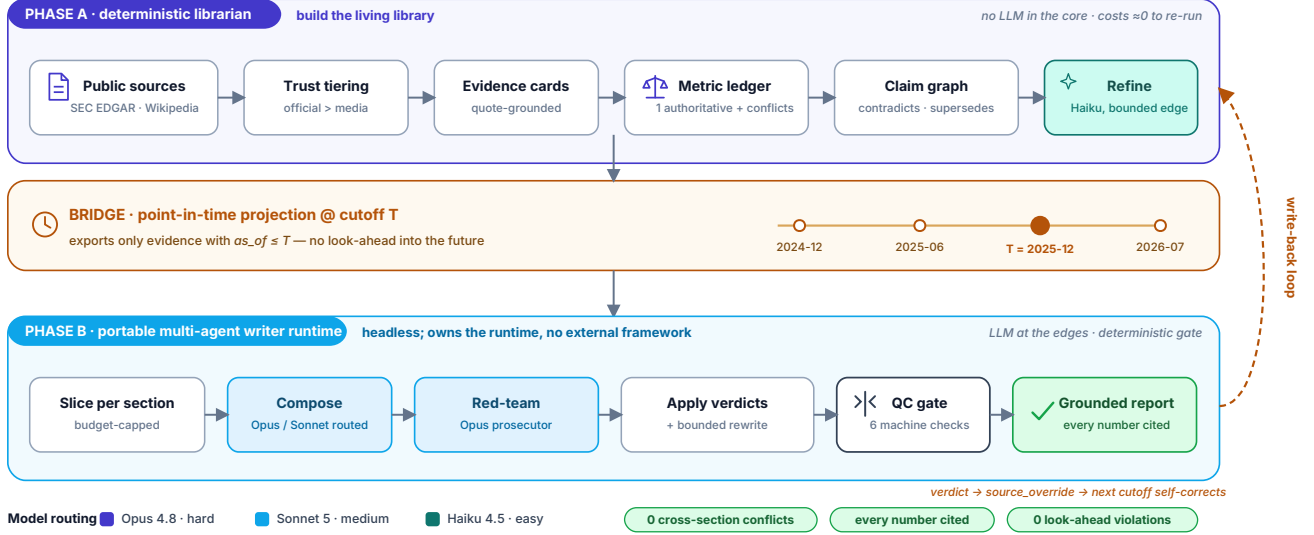


Figure 1: End-to-end system. Phase A (deterministic librarian) ingests timestamped public sources into trust-tiered evidence cards, a metric ledger, and a claim graph. The cutoff dial selects a knowledge time T ; only evidence with $as_of \leq T$ is exported. Phase B (portable agent runtime) slices per section, composes with difficulty-tiered model routing, red-teams, and passes a deterministic QC gate. Red-team verdicts feed back to the store.

Algorithm 1: Maintain library, write report @ T .

```

input : public sources  $S$ ; cutoff  $T$ ; outline  $O$ 
output: grounded report  $R$ ; updated store  $\mathcal{L}$ 
// Phase A: deterministic librarian (LLM-free core)
foreach  $s \in S$  do
    tier,  $as\_of \leftarrow \text{classify}(s)$ 
    cards  $\leftarrow \text{extract quote-grounded evidence from } s$ 
    if numeric & ambiguous then Haiku-refine card vs. its quote
ledger  $\leftarrow \text{per (co., metric): pick by tier} \rightarrow \text{corrob.} \rightarrow \text{recency}$ 
graph  $\leftarrow \text{contradicts / supersedes / qualifies edges}$ 
// Bridge: point-in-time projection (no look-ahead)
 $E_T \leftarrow \{c \in \mathcal{L} : as\_of(c) \leq T\}$ ; recompute ledger @  $T$ 
// Phase B: portable multi-agent writer
foreach section  $\sigma \in O$  (bounded parallel) do
     $C_\sigma \leftarrow \text{Slice}(E_T, \sigma)$  capped by salience budget
     $d_\sigma \leftarrow \text{Opus if } \sigma \text{ touches unresolved conflict else Sonnet}$ 
    draft $_\sigma \leftarrow \text{normalize(Compose}(C_\sigma; \text{model } d_\sigma))$ 
     $v_\sigma \leftarrow \text{RedTeam(draft}_\sigma)$  // Opus prosecutor
apply verdicts; bounded rewrite; deterministic convergence
backstop
while  $\text{QC}(R) \neq \emptyset$  and rounds < cap do
    rewrite flagged sections
foreach refuted  $v_\sigma$  do WriteBack( $\mathcal{L}$ ) // source_override

return render( $R$ ),  $\mathcal{L}$ 

```

may inform entity/topic routing and context but can never become a citable hard-evidence number, and a supporting-tier macro value can never displace a company’s own official figure: the “official-first” rule.

Metric ledger. For each (company, metric) the ledger selects one authoritative value by a fixed policy: *tier* dominates, then *corroboration* (distinct-source count) breaks ties within a tier, then *recency* (as_of). Competing values are retained as alternatives and flagged as a conflict when a comparable same-kind figure materially disagrees (a fixed $> 15\%$ threshold, so unit-equal restatements do not spuriously fire), but the report cites the single authoritative value, which is what eliminates cross-section drift.

QC gate (six deterministic checks). The delivery gate is language-neutral and LLM-free: (1) orphan citations, (2) unsourced numbers, (3) numeric drift across sections, (4) buried contradictions (a claim-graph conflict whose two endpoints are not reconciled together), (5) unregistered metrics, (6) cross-section contradiction. A report is deliverable only when the error set is empty (the six checks are specified in Appendix B). Because this gate computes the headline metrics, we validate the gate itself (§5, E5).

4 Deployment & Dataset

We self-collected a public, redistributable, English-only corpus at *production scale* (Table 1): **6,130 sources** extracting to **555,926 evidence cards** (457,561 numeric) and a metric ledger of 2,589 authoritative company-metric values, 2,132 of them carrying recorded

Table 1: Self-collected corpus (public, timestamped, multi-tier, multi-sector).

Property	Value
Sources registered (total)	6130
official (SEC filings, hard evidence)	5397
gov_stat (BLS macro, supporting)	672
media (Wikipedia, routing only)	61
of which produced ≥ 1 card	6054
Sectors covered	11
Companies (issuers)	295
Evidence cards	555926
Numeric cards	457561
Media→numeric leakage	0
Publication span	2023-06 – 2026-07

conflicts. The three tiers are 5,397 SEC EDGAR filings [23] (official) for 295 issuers across 11 sectors, 672 U.S. Bureau of Labor Statistics (BLS) macro releases (gov_stat: authoritative supporting context that, carrying no company attribution, by design never enters the per-company ledger), and 61 Wikipedia articles (media: routing and context only, by design backing no numbers). Filings are fetched keyless via the official EDGAR and BLS REST APIs with provenance headers.

The design point is **one library, many reports**: a single maintained store serves multiple report theses rather than being purpose-built for one. From this store we generate four flagship point-in-time reports at a common cutoff (2025-12-31) (AI-compute, energy, healthcare/pharma, and banks), each projected by the sector-scoped bridge, and each passing the QC gate with zero errors. Breadth matters for the evaluation too: the corpus contains thousands of naturally-occurring cross-period and cross-source contradictions (drift in remaining performance obligations, backlog, and capital expenditure; restatements; revised guidance) across every sector, never fabricated drift. Each source is retained under its issuing API’s terms, with per-tier licensing recorded.

Because every source carries its true publication date, we replay time by filtering the store to a cutoff (Phase B), mimicking “30/60/90 days later” without re-fetching: the production case where later filings revise or contradict a value an earlier report relied on.

5 Evaluation

Every headline metric is machine-computed (no LLM decides a reported number) and, where an ablation applies, compared on the *identical* corpus against an explicit baseline arm. Deterministic experiments (E1, E3, E4, E5, E7, E8) are exact by construction; the LLM-dependent cost/latency/quality comparison (E6) is run on the real Bedrock backend over three repeats. All numbers in the tables are emitted directly from the stored per-experiment result records (reproduction steps in Appendix G); the headline results are also charted together in Fig. 4 (App. E).

E1: a shared ledger removes cross-section drift. Without a shared ledger, a writer grounding each section independently surfaces every competing value for a metric; the ledger collapses each to one authoritative value. On the real store at the final cutoff, the no-ledger baseline would emit **6,845** contradictory figures across 2,105

metrics with competing values; ours emits **0** (Table 2). Replayed across seven cutoffs, the ledger’s authoritative value changed **4,732** times, and *every* change was justified by newer, higher-tier, or more-corroborated evidence (0 unexplained).

E2: grounding. Across all four flagship theses composed from the one library (AI-compute, energy, healthcare/pharma, banks; cutoff 2025-12-31), every numeric-bearing body line must carry an evidence citation or a metric annotation. Aggregate grounding is **202/203 (99.5%)** with **0** orphan citations and **0** unregistered metrics; three reports are 100%, and the lone exception is a synthesis sentence whose figures are each cited earlier in the same section.

E3: trust tiering suppresses rumor and quarantines macro context. End-to-end on the real store, all three tiers classify correctly (5,397 filings as hard_evidence, 672 BLS releases as supporting_evidence (gov_stat), 61 Wikipedia articles as routing_only, **0** misclassified), and **0** of the 457,561 numeric cards trace to a routing-only source. The third tier is genuinely mined (2,352 numeric gov_stat cards: CPI, PPI, payrolls, unemployment), yet because a macro statistic carries no company attribution, **0** gov_stat values displace a company’s own official figure and the per-company ledger stays 100% official: neither media nor macro statistics ever become a citable number, and the invariant holds across the full production-scale corpus (all 457,561 numeric cards, 295 issuers, 11 sectors).

E4: tier-first selection beats popularity. We run two selection policies on one labeled gold set of metric clusters: our tier-first ledger, and a popularity-first baseline that takes the value with the most distinct backing sources (the “most-cited”/semantic-layer heuristic), ignoring tier. Tier-first is correct on **22/22** cases; popularity-first scores only **9/22** (Table 3). Thirteen cases are *popularity traps*: a widely-repeated lower-tier value (a rumor echoed by several media sources, or a corroborated macro statistic) competes with a single official filing; the tier rule survives all thirteen, popularity adopts the wrong value every time. The set spans the full configured lattice (official > gov_stat > sell_side > media), including the invariant that a newer, more-corroborated gov_stat value still cannot displace an official figure (which gov_stat may anchor only when none exists), and corroboration breaking ties only *within* a tier. It is a *designed coverage lattice*, not a sample: the cross-tier conflict it probes cannot arise on this corpus (all **2,589** real ledger clusters are single-tier, as gov_stat cards carry no issuer name and media is never mined for a number), yet the within-tier rule is exercised at scale: over **2,132** real conflicts our corroboration→recency policy differs from naive newest-wins on **973**, far beyond the 22 gold cases.

E5: the checker is trustworthy (recall and precision). A gate is only trustworthy if it both catches real defects and stays quiet on clean text. We clone a clean, QC-passing run and (i) inject five defect classes (orphan citation, unsourced number, broken cross-reference, unregistered metric, buried contradiction) and (ii) apply three *negative controls*: defect-free perturbations that must *not* fire (a paragraph reusing only already-valid citations and metric tokens, a duplicated grounded line, number-free prose). Recall is **1.0** (5/5) and precision is **1.0** with a **0** false-positive rate on the controls (Table 4). Crucially, we separate delivery-blocking from advisory detection: the delivery gate is “error set empty,” and 3/3 error-level defects raise a blocking error, while the two warning-level defects

are caught but advisory by design. This disarms “graded your own homework”: the gate that computes E1–E4 is itself validated on both axes.

E6: in the distributed writer, parallelism and routing cut cost at comparable quality. Isolating the multi-agent compose fan-out on identical slices over three repeats (Table 5), bounded-concurrency parallelism across the per-section agents runs $3.7\times$ faster than serial, and difficulty-tiered routing (conflict-touching sections $\rightarrow$ Opus, the rest $\rightarrow$ Claude Sonnet 5) costs **4.1% less** than sending every section to Opus. The cost gap is deliberately modest on this flagship: it is conflict-heavy, so 5 of 6 sections touch an unresolved edge and correctly route to Opus, so routing saves little precisely when the report is hard, and the same dial saves far more on a low-conflict thesis. All four variants pass QC with zero errors, but a binary gate cannot rank them, and all-Sonnet is the cheapest, so “equal quality” needs an independent signal. We add a deterministic, graded quality score (grounding coverage + conflict-pair coverage + output-contract adherence, computed on each variant’s actual prose, independent of the QC gate). The counterintuitive result: tiered routing scores **above** the all-Opus ceiling (+0.079) and far above all-Sonnet (+0.262). Spending the strong model only where reasoning is hard beats spending it everywhere, so the cheaper all-medium point is not the default: it saves on easy sections but degrades exactly the conflict-heavy synthesis that routes to Opus.

E7: the living library grows without look-ahead. Replaying the store across seven cutoffs (Table 6, Fig. 2) yields **0** look-ahead violations and monotonic growth (235,373 $\rightarrow$ 555,312 cards; 1,659 $\rightarrow$ 6,054 card-bearing sources), capturing **4,395** post-initial evidence-arrival events (new filings, restatements, new conflicts) and lifting recorded metric conflicts from 1,770 to 2,132. This closes end to end at the report level: regenerating the AI-compute flagship at three advancing cutoffs yields a report that grows in lockstep (27,104 $\rightarrow$ 42,900 cards, 255 $\rightarrow$ 276 metrics, 562 $\rightarrow$ 655 reconciled conflict edges) while every cutoff stays deliverable (QC errors = 0), and all four flagship theses compose at the shared 2025-12-31 cutoff from this one library. A static one-shot corpus cannot exhibit this: as real evidence arrives, the library grows and corroboration rises.

E8: the write-back loop self-corrects a later run. Growth is only half of “living”; the loop must also close on *corrections*. We trace one worked case end to end using the librarian’s real override machinery (Table 7, App. F). In this illustrative scenario the report-side red-team challenges the authoritative interest figure (\$19mn) after flagging its backing filing as low-confidence. The verdict maps to a librarian source_override (status retracted); on re-ingestion the 284 evidence cards from that filing inherit the non-active status, and because the metric ledger considers only active-source cards, the authoritative value self-corrects to \$9mn, an already-recorded alternative, with **0** manual value edits. Tracing the same override path on a batch of auto-discovered conflicted metrics, 5 of 6 write-backs self-correct, each to a pre-existing same-kind alternative (0 manual edits), so the loop closes on many values, not one, deterministically and auditably.

Limitations. The corpus is English-only and three-tier as collected (official SEC filings, gov_stat BLS macro statistics, media Wikipedia; sell_side omitted, §3). The gov_stat tier is authoritative

Table 2: E1: cross-section figure drift (lower is better).

Condition	drift figs	metrics w/ conflict	justified drift
Baseline (no shared ledger)	6845	2105	—
Ours (shared ledger)	0	2105	—
Temporal drift, 7 cutoffs: 4732 changes, 100% evidence-justified, 0 unexplained.			

Table 3: E4: authoritative-value selection vs. gold.

Gold case	ours	popularity-first
official beats 3 $\times$ media rumor ^f	✓	✗
official beats newer 2x sell ^{†t}	✓	✗
newer official wins tie	✓	✓
corrob tie-break	✓	✓
sell beats 2x media ^{†t}	✓	✗
corrob > newer single	✓	✓
gov beats 3 $\times$ media rumor ^f	✓	✗
official beats newer gov ^f	✓	✗
official beats 2 $\times$ gov ^f	✓	✗
gov fills absent official	✓	✓
gov corrob tie-break	✓	✓
newer gov revision wins	✓	✓
official beats 4 $\times$ media rumor ^f	✓	✗
official beats 3 $\times$ sell ^{†t}	✓	✗
gov beats 2 $\times$ media ^f	✓	✗
modal guards \$-misparsed	✓	✓
gov beats 2 $\times$ sell ^{†t}	✓	✗
official > gov > media (lattice) ^f	✓	✗
newer official on equal corrob	✓	✓
kind guard drops stray percent	✓	✓
official beats 5 $\times$ media crowd ^f	✓	✗
official beats corrob newer gov ^f	✓	✗
Accuracy	22/22	9/22

ours = tier-first (ledger policy). ^f 13 popularity traps (widely-repeated lower-tier value); tier rule survives all 13. [†] configured-but-unused tier.

Table 4: E5: QC gate meta-evaluation, recall and precision (defect injection plus negative controls).

Perturbation	level	QC outcome
orphan citation	error	caught, blocks delivery
unsourced number	warning	caught, advisory (warning)
broken xref	warning	caught, advisory (warning)
unregistered metric	error	caught, blocks delivery
buried contradiction	error	caught, blocks delivery
neg: reuse existing citations	(clean)	0 new errors
neg: duplicate grounded line	(clean)	0 new errors
neg: prose only no numbers	(clean)	0 new errors
Recall 5/5=1.00, precision 1.00, FP-rate 0.00		
over 3 controls; 3/3 error-level defects block delivery (rest advisory).		

but macro-only: its releases carry no company attribution, so by construction it enriches context, not the per-company ledger. Entity linking is substring-based, so it occasionally over-attributes a metric when a company’s short name is a substring of unrelated

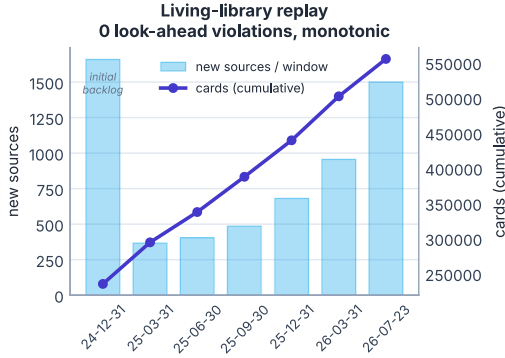


Figure 2: E7: point-in-time replay across seven cutoffs. Bars (left axis) count sources newly available in each window; the line (right axis) is the cumulative evidence-card base, which grows monotonically with zero look-ahead (a report as of T reads only $as_of \leq T$). The first bar is the pre-existing back-catalog loaded at the opening cutoff, not a per-window rate.

Table 5: E6: cost/latency of routing and parallelism (compose fan-out, real Bedrock).

Configuration	mix	\$/rep	wall (s)	quality
All-hard (Opus), par.	6H/0M	\$2.35	64	0.714
All-medium (Sonnet), par.	0H/6M	\$0.49	47	0.531
Tiered, serial	5H/1M	\$2.26	184	0.801
Tiered, par. (ours)	5H/1M	\$2.25	50	0.793

Mean over 3 repeats. Tiered vs all-hard: 4.1% cost saved, $3.7\times$ faster than serial. Quality (grounding+conflict+contract, indep. of QC): tiered-all-hard = +0.079, -all-medium = +0.262; all pass QC.

Table 6: E7: living-library replay across cutoffs.

Cutoff T	cards	sources	metrics	conflicts	new src
2024-12-31	235373	1659	2279	1770	1659
2025-03-31	294477	2026	2343	1868	367
2025-06-30	337597	2430	2388	1921	404
2025-09-30	387741	2916	2462	2001	486
2025-12-31	439605	3597	2493	2026	681
2026-03-31	502134	4553	2544	2093	956
2026-07-23	555312	6054	2589	2132	1501

0 look-ahead violations; monotonic. New-src sums to 6054 card-bearing sources; 4395 post-initial arrivals (excl. the 1659 at the first cutoff).

filing text (e.g. “3M”); this is an auditable extraction artifact, not a ledger-policy error. The E4 cross-tier gold set is designed rather than sampled (the deployed corpus has no cross-tier clusters), though the within-tier rule is corroborated on 2,132 real conflicts; E6 uses a deterministic quality proxy, not human judgement; and E8 traces write-back on a small batch. E1’s no-ledger arm isolates the ledger’s effect, not a strong shared-state competitor (a graph- or semantic-layer-backed retriever), the natural next comparison our design is built to host.

6 Lessons & Related Work

Lessons. (i) *Govern by trust, not popularity*: a most-cited-value heuristic adopts widely-repeated rumor, while an official-first ledger is simpler and correct (E1, E3, E4). (ii) *Keep the core deterministic, put the LLM at the edges*: deterministic selection, consistency, and QC make the headline metrics reproducible and the gate meta-evaluable (E5). (iii) *Own the runtime*: standard multi-agent building blocks (file-per-agent artifacts, contract-first outlines) carry over cleanly to a headless service without binding to any external agent framework. (iv) *Routing buys quality, not just cheapness*: tiered composition matches or beats the all-Opus ceiling while a uniform-cheap baseline degrades, at modest dollar saving on a conflict-heavy report (E6).

Related work. Grounding and citation. RAG grounds individual answers [10, 16] per query rather than maintaining a reconciled ontology, enforcing no cross-time consistency; Self-RAG [3] adds a self-critique, the same model judging its own output, not a deterministic, meta-validated gate. Grounding metrics like FActScore [19] and RAGAs [8], citation benchmarks like ALCE [9], and rubrics like FinReasoning [30] *score* whether a text is supported but do not *govern* a living store; our QC gate blocks delivery, not a post-hoc score. STORM [21] synthesizes Wikipedia-like articles from retrieved sources but has no source-authority tier, metric ledger, or point-in-time discipline.

Temporal and streaming settings. Closest is *temporal/streaming QA*: TimeQA [4], StreamingQA [17], and RealTime QA [14] isolate look-ahead and evolving knowledge, but for *short-answer* questions; we target long-form reports whose challenge is *cross-section* consistency. GraphRAG [7] and GFM-RAG [18] reason over an entity/claim graph but carry no provenance-tier or point-in-time projection, reconciling per query not in a maintained store; MemGPT [20] persists agent state without source-authority or no-look-ahead discipline.

Agentic frameworks and verification. Agentic frameworks [11, 22, 24, 25] and toolkits such as LangGraph [15] provide the stateful multi-actor graph we deliberately *do not* bind to, building a minimal headless orchestrator of our own instead (Lesson iii). Verified orchestration [27] closes a plan-verify-replan loop like ours, but we keep the trust backbone and delivery gate deterministic and meta-validated (E5), with human-gated promotion via an append-only log (E8). Registry-driven grounding like REGAL [1] and ontology-guided extraction like OntoMetric [26] share our deterministic-core discipline but ground structured telemetry or a single-document ESG graph, not *unstructured, cross-document* evidence reconciled into a timestamped ledger. Financial QA benchmarks [5, 12] and FinCARDS [29] target intra-document reasoning; semantic-layer analytics [6] share the “governed source of truth” intuition we extend to timestamped, trust-tiered evidence, with LLM-as-judge [28] used only as a red-team.

7 Conclusion

Separating a maintained, trust-tiered, point-in-time library from report writing turns long-form generation’s chronic drift, provenance loss, and trust flattening into mechanically-checkable, largely eliminated properties.

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A Worked artifact: one metric, end to end

We trace a single real metric, Oracle’s Remaining Performance Obligations (RPO), through the three librarian artifacts and into the delivered report, to make the data model concrete. All snippets are verbatim from the store and a delivered run (identifiers abbreviated for space).

(A) *Evidence card*. The deterministic extractor emits one quote-grounded card per (source, metric) hit. The card carries the verbatim quote, so every downstream number is auditable back to the filing text:

```
{ "evidence_id": "ev_orcl_..._rpo_d0c7b8b4",
  "source_id": "src_orcl_..._10_k",
  "company": "Oracle", "metric": "RPO",
  "quote": "Remaining performance obligations were
    $638 billion and $138 billion as of May 31,
    2026 and 2025, respectively.",
  "metric_value": "$638 billion", "value_norm": 638000.0,
  "value_kind": "money_mn", "source_tier": "official",
  "as_of": "2026-06-22", "evidence_kind": "quantitative",
  "source_status": "active" }
```

Listing 1: Evidence card (SEC 10-K, official tier).

(B) *Metric ledger*. For each (company, metric) the ledger selects one authoritative value by tier → corroboration → recency and retains the losers as alternatives with a disagrees flag: this is what a section cites, and what makes cross-section drift impossible:

```
{ "metric_id": "mtr_oracle_rpo", "company": "Oracle",
  "metric": "RPO", "authoritative_value": "$638.0bn",
  "value_norm": 638000.0, "as_of": "2026-06-22",
  "source_tier": "official", "value_conflict": true,
  "basis_evidence_id": "ev_orcl_..._rpo_d0c7b8b4",
  "alternatives": [
    { "value": "$552.6bn", "as_of": "2026-03-11", "disagrees": false },
    { "value": "$523.3bn", "as_of": "2025-12-11", "disagrees": true },
    { "value": "$455.3bn", "as_of": "2025-09-10", "disagrees": true } ] }
```

Listing 2: Metric-ledger row with retained alternatives.

(C) *Delivered report prose*. The composed section writes numbers not as literals but as symbolic ledger handles (e.g. {m_oracle_rpo: authoritative}) that are substituted at render, and cites evidence-card ids inline ([E14847]), so no number is typed by hand and every one is traceable. The rendered text reads:

“As of May 31, 2026, Oracle reported remaining performance obligations (RPO) of \$638.0bn, up sharply from \$455.3bn three quarters earlier [E14847][E14761]. ... Oracle expects to recognize only about 10% over the next twelve months [E14847].”

(D) *Claim-graph edge*. Cross-source and cross-period tensions are stored as typed edges; a supersedes edge is what lets a later filing override an earlier value while both stay auditable:

```
{ "edge": "supersedes", "metric": "RPO", "company": "Oracle",
  "from_evidence": "ev_orcl_..._rpo_d0c7b8b4", // $638.0bn @ 2026-06-22
  "to_evidence": "ev_orcl_..._rpo_e1366a04", // $455.3bn @ 2025-09-10
  "reason": "newer official filing, same metric" }
```

Listing 3: Claim-graph edge (supersedes).

B The QC gate (six deterministic checks)

The delivery gate is language-neutral and LLM-free; a report is deliverable only when the error set is empty. Checks (1) and (4)–(6) raise blocking errors; (2) is advisory (warning-level). Check (3)

cannot fire once one authoritative value backs every annotation (the drift E1 removes structurally). E5 validates the gate by defect injection.

- (1) **Orphan citation**: a citation marker with no backing evidence card.
- (2) **Unsourced number**: a numeric body line with neither an evidence citation nor a metric annotation.
- (3) **Numeric drift**: the same metric rendered with two different values across sections (the drift E1 eliminates).
- (4) **Buried contradiction**: a claim-graph conflict whose two endpoints are not reconciled in the same place.
- (5) **Unregistered metric**: a metric handle cited but absent from the ledger.
- (6) **Cross-section contradiction**: mutually inconsistent statements across sections.

C Ontology schemas

Both tiers speak the same three objects, which is why the bridge is a data adapter rather than a rewrite. Required fields (from the JSON-Schema definitions the two tiers share):

- **EvidenceCard**: evidence_id, project_id, source_id, fact, source_tier, confidence; quote required when allowed_use=hard_evidence.
- **MetricLedger row**: metric_id, company, metric, authoritative_value, value_norm, source_tier, basis_evidence_id, as_of, alternatives.
- **Claim**: claim_id, subject, predicate, object, supporting_evidence, contradicting_evidence.

D The distributed writer, illustrated

Figure 3 draws the four design points of §2.1 as one picture: the outline is a partition whose sections a difficulty router assigns to a model tier (heterogeneous agents); the sections fan out across a bounded-concurrency pool of composer/red-team pairs (parallelism and separation of powers); and the agents coordinate only through the shared trust-tiered store (read at compose time, written back by the red-team) rather than by messaging each other (stigmergic coordination). The single authoritative value lives in the store, so no two sections can cite different numbers for the same metric even while they run concurrently.

E Results at a glance

Figure 4 visualizes the headline quantitative results whose exact figures are tabulated in §5, on a 2×3 grid: tier-first vs. popularity-first selection (E4), QC defect-injection recall and precision (E5), ledger growth with fully-justified drift (E1), and the E6 cost, latency, and graded-quality trade-off across routing variants.

F Write-back trace

Table 7 traces the one worked write-back correction of E8 (§5) step by step: a red-team refutation maps to a librarian source override that self-corrects the ledger’s authoritative value at the next cutoff.

The distributed writer: heterogeneous section agents over one shared store

The outline is a partition; sections fan out to model-routed composer/prosecutor pairs that coordinate only through the shared store — never with each other.

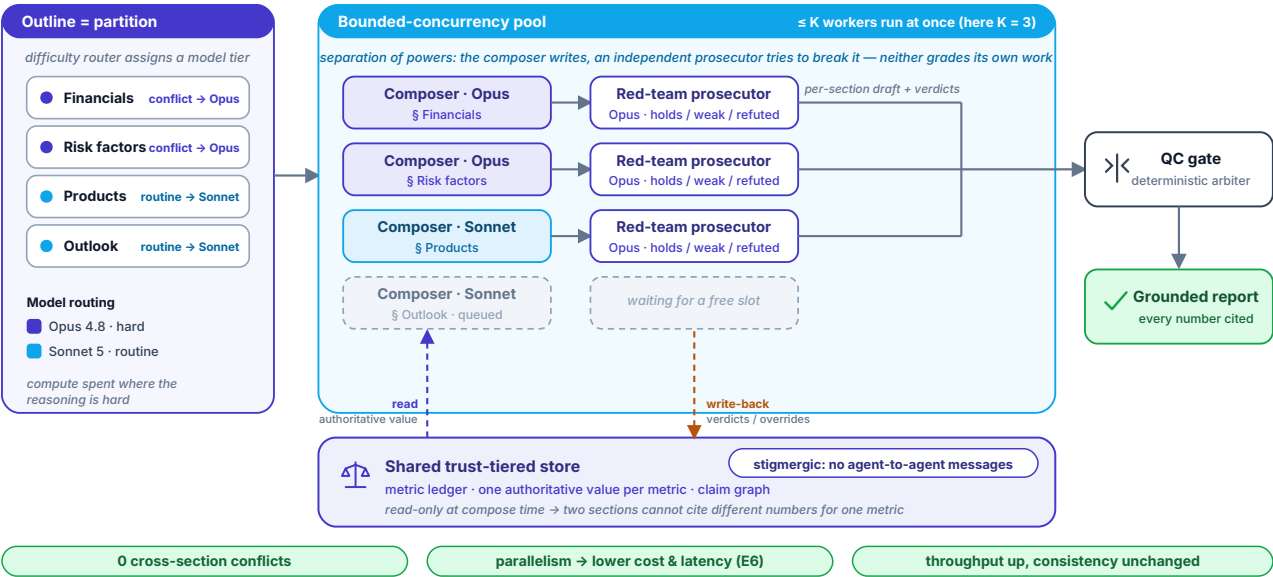


Figure 3: The distributed multi-agent writer (Phase B, §2.1). The report outline is partitioned into sections; a difficulty router sends conflict-touching sections to Claude Opus 4.8 and routine ones to Claude Sonnet 5. Sections fan out across a bounded-concurrency pool ($\leq K$ workers), each a composer paired with an independent Opus red-team “prosecutor” that cannot grade its own work. Agents never message each other: they coordinate *stigmurgically* through the shared trust-tiered store (reading the single authoritative value per metric and writing verdicts back), so parallelism never costs cross-section consistency. A deterministic quality-control gate, not an LLM, is the final arbiter.

Table 7: E8: a red-team refutation writes back and self-corrects the ledger.

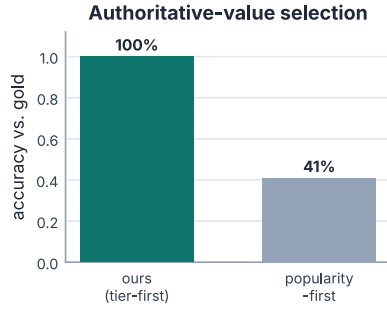
Write-back step	state
Metric (worked case)	issuer X, interest
Authoritative value (before)	\$19mn
Red-team verdict	refuted on backing filing
Librarian action	source_override: retract; 284 cards re-stamped
Authoritative value (after)	\$9mn (prior alternative)
Metrics self-corrected (batch)	5 of 6 traced
Manual value edits (all cases)	0

G Reproducibility

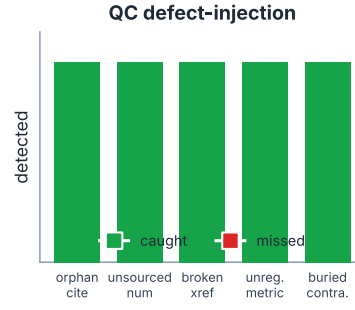
Deterministic experiments (E1, E3, E4, E5, E7, E8) run with no cloud and reproduce exactly from a fixed store; E6 hits the real Bedrock backend. Every table and figure is emitted programmatically from the stored per-experiment result records. The deterministic experiments run against an offline stub backend, and each source’s per-tier licensing and fetch parameters are recorded with the stored corpus.

H Ethical, legal, and societal considerations

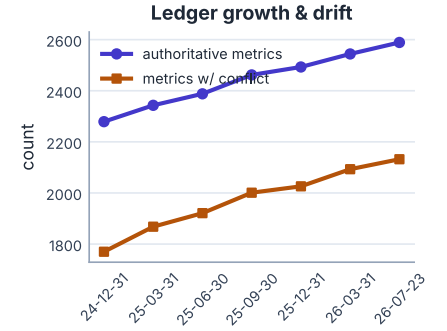
The system operates only on public, timestamped financial sources used under each source’s terms (with per-tier licensing documented): it ingests and reports metrics but never emits any customer, personal, or otherwise non-public data, and every delivered figure is traceable to its source, so provenance is auditable rather than obscured.



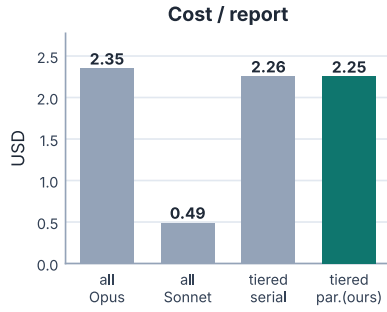
(a) E4: selection accuracy vs. gold (22/22 vs. 9/22)



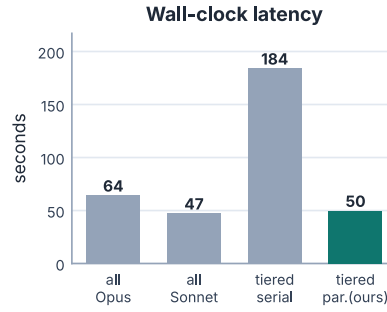
(b) E5: QC recall & precision (1.00 / 1.00)



(c) E1: ledger growth & drift (100% of 4,732 justified)



(d) E6: cost / report (USD)



(e) E6: wall-clock latency (s)



(f) E6: graded quality (0–1)

Figure 4: Headline results as vector charts (companion to Tables 2–5), on a 2×3 grid: top row, the consistency and evaluation results (E4/E5/E1); bottom row, the E6 cost/latency/quality trade-off across routing variants (ours in teal: cost at the all-Opus level, $3.7\times$ faster than serial, quality above the ceiling).